



PHN 300: Lab-Based Project

(Supervisor: Prof. Rajesh Kumar)

Topic - *Non-volatile switches based upon $\text{Ge}_2\text{Sb}_2\text{Te}_5$ embedded SOI channel waveguides for operation at 2.1 μm*

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B. Tech. (Engineering Physics)



Acknowledgement

I sincerely appreciate the guidance and advice given by Prof. Rajesh Kumar. He guided me during this study with apt. resources, encouragements, and methodologies. I also express my thanks to Nadir Ali, who helped me with the simulations, & trained me to construct models using CST MW Studio.



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Abbreviations Used

GST – $\text{Ge}_2\text{Sb}_2\text{Te}_5$

SOI – Silicon-on-Insulator

IL – Insertion Loss

ER – Extinction Ratio

EO – Electro-Optic

1x1 – one input-one output

ITO – Indium-Tin Oxide

PCM – Phase Change Material

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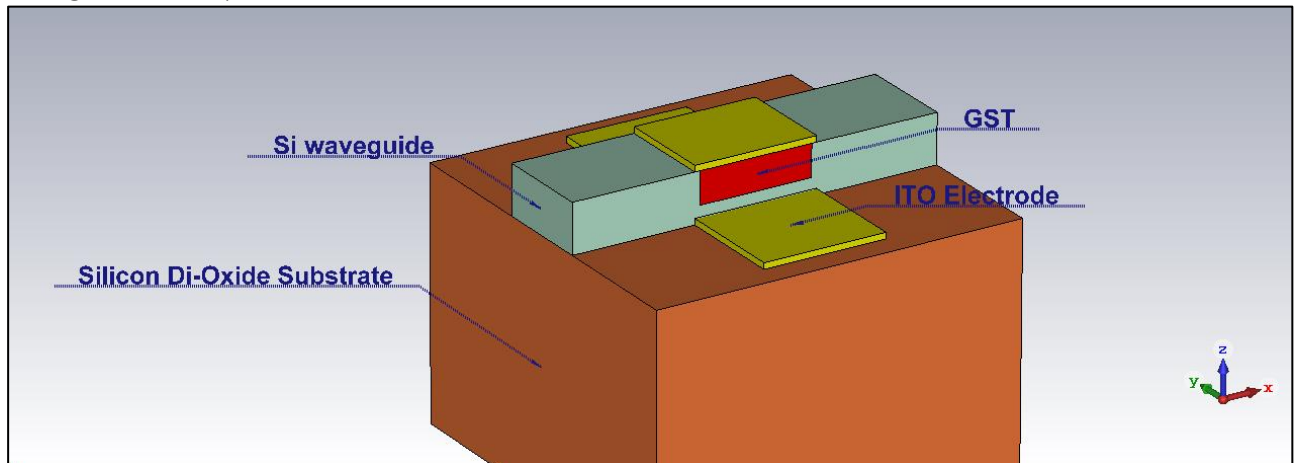
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1. Abstract

This report is a study (using simulations) of one of the many theoretical designs of electro-optical 1x1 high static performance on-off switch based upon GST-embedded SOI channel waveguides for operation in the mid-infrared wavelength of 2.1 μm . The state of the switch can be changed using electrical/optical pulses, and no electrical bias is needed to maintain a particular state. Because the active material (GST) has two “stable” phases, i.e., crystalline and amorphous, the device holds itself in either of the two states, and the voltage/optical pulse needs to be applied only during transition. In this project, the reversible transition between the switch states is achieved by inducing phase transition from crystalline to amorphous and vice-versa by application of voltage pulses using ITO as the electrode. The approach of embedding the nanoscale active material GST within the Si waveguide is taken to enhance the interaction of light with the active region of the switches. The dimensions of the active regions of the switches are optimized to achieve low insertion loss, low switching energy, and high extinction ratio. Doping of Si beneath the GST is necessary to facilitate the electrical conduction needed for Joule heating. It is assumed that the amorphous and crystalline phases of GST can be triggered electrically by Joule heating current applied to GST sitting on top of the partially etched doped-Si waveguide. This project was a sincere effort to replicate the results of the research already conducted by Mr. Nadir Ali & Prof. Rajesh Kumar and to further extend the analysis by testing a variation of all the parameters based on the concentration of doping in the layer of Si beneath the GST.

(Keywords: Nano photonic devices, photonic switches, Mid-IR optical devices, silicon photonics, Phase change materials)





2. Software Used

The software used in the simulations is called “CST Design Environment”. This powerful software contains diverse functions for simulating electromagnetic problems and is made up of six independent functional studios:

1. CST Microwave Studio,
2. CST EM Studio,
3. CST Particle Studio,
4. CST Design Studio,
5. CST PCB Studio and
6. CST Cable Studio.

CST Microwave Studio is a specialized tool for the fast and accurate 3D EM simulation of high-frequency problems. Along with a broad range of applications, CST Microwave Studio offers considerable product-to-market advantages: shorter development cycles, virtual prototyping before physical trials, and optimization instead of experimentation. All the simulations in this report were accomplished using the CST Microwave Studio. Microwave Studio is a functional module that can deal with complex high-frequency problems and has the following advantages -

Firstly, it contains the transient solver for efficient calculation of loss-free and lossy structures. The solver does a broadband calculation of S-parameters from one single calculation run by applying DFTs to time signals. The frequency-domain solver can process adaptive sampling and has an expert system based automatic mesh generation with 3D adaptive mesh refinement. The general-

purpose solver supports both hexahedral and tetrahedral meshes. Besides the general-purpose solver, the frequency domain solver also contains two specialized solvers for strongly resonant structures (for use with hexahedral meshes only). The first of these solvers only calculates S-parameters, whereas the second one also calculates fields and, therefore, requires some additional calculation time.

Secondly, it can calculate 3D Eigenmodes. The structure can be viewed either as a 3D model or as a schematic. The latter allows for easy coupling of the EM simulation with circuit simulation. It contains a fast and memory-efficient FI-method. The FIT (Finite Integration Technique) method that the studio uses for simulation performs exceptionally well due to the Perfect Boundary Approximation (PBA) for solvers using hexahedral grids. The transient and Eigenmode solvers also support the Thin Sheet Technique (TST). All solvers support hexahedral grids.

Thirdly, it contains an advanced ACIS based, parametric solid modeling front-end function with excellent structure visualization. This means that simple modeling can be directly accomplished in the studio, and a feature-based hybrid modeler allows for quick structural changes. Meanwhile, it can also import or export models of all kinds of file types; for example, SAT, Step, Autodesk Inventor, VDA-FS CATIA, Pro/E, IGES, NASTRAN, or STL 3D CAD data, and DXF 2D CAD data. It can also import GDSII and Gerber 2D CAD data.

3. Introduction

3.1 Motivation

Integrated photonic circuits are the way forward for future communication systems that have the potential to overcome the limitations that the electronic communication systems are facing currently, such as latency between processor and memory, high power consumption, and limited bandwidth. Conventional copper interconnects limit the speed and cross-talk increases as the density of interconnects increases. Silicon photonics platform is the leading candidate for inter- and intra-chip communication primarily due to its low cost and high yield fabrication availability because it can take advantage of highly developed CMOS manufacturing process used in the Silicon (Si) microelectronics industry. Silicon photonic switches are in high demand to realize Si-based photonic integrated circuits as they offer high modulation speed and picosecond switching time. For on-chip photonic devices and integrated circuits, phase change materials (PCMs) are a suitable candidate because of their non-volatile nature. PCMs have several prominent properties such as high optical contrast between amorphous and crystalline states, phase transition time of sub nanoseconds, compatibility with the CMOS manufacturing process, and high scalability, which make them ideal candidates for chip-scale active photonic devices. PCMs like GST can be switched rapidly and repeatedly between amorphous and crystalline phases by the application of short electrical/optical pulses. A fully functional next-generation photonic system will require a full range of

suitable components with higher integration, miniaturization, and complex functionalities.

3.2 Phase Change Material: GST

The phase-change material $\text{Ge}_2\text{Sb}_2\text{Te}_5$, commonly known as GST, is reported to have optical response superior to that of the other opto-electronically attractive phase-change material vanadium dioxide (VO_2). The active operation in the integrated photonic devices is enabled because GST exhibits good phase change behavior between its amorphous (a-GST) and crystalline (c-GST) phases. The transition between two phases of GST is rapid, repeatable, and results in a huge difference in optical and electronic properties. The phase change material GST exhibits a drastic change in its real as well as the imaginary part of the refractive index ($\Delta n > 1$ and $\Delta k \sim 0.4$) when switched between its two stable phases, leading to a substantial change in optical phase and amplitude within a small active length. Because of the high refractive index difference between the two phases of GST, active devices exhibit a high on-off ratio and can have broadband operation with an ultra-compact footprint. Another interesting property of GST-based devices is the self-holding bi-stability i.e. no continuous power is required to maintain the device state. The self-holding bistability in GST at room temperature enables a low energy consuming platform for non-volatile photonic components. The switching of GST from the a-GST to c-GST phase and vice-versa is achieved by Joule heating of the GST material through the application of optical and/or electrical pulses. These transitions occur with a transition time of sub-nanoseconds. Moreover, the fabrication of GST-based devices is also CMOS compatible. Hence, by using these characteristics of GST combined with Si waveguides, ultra-small energy-efficient, and non-volatile optical switches can be realized.

3.3 Wavelength of operation: 2.1 μm

There is a special interest in switching near wavelengths of 2000 nm because a new generation of hollow-core photonic bandgap fibers (HC-PBGF) is being developed for operation in the 1900 to 2100 nm wavelength band and such fiber transmission is central to increasing the information capacity of global fiber-optic networks. Internet data traffic over the fiber optic network has been growing at an exponential rate and is rapidly approaching the projected ultimate data transmission capacity limits, raising concerns of a future capacity crunch. The HC-PBGFs guide light in the air core, which minimizes Rayleigh scattering loss by having a high wavelength of operation as Rayleigh scattering has λ^{-4} dependence. Such fibers have a minimum loss window that falls around the 2 μm region. Additionally, HC-PBGFs exhibit low nonlinearity and low latency compared to conventional fibers. A new, non-conventional physical technology will soon be indispensable. Combining HC-PBGF with a Thulium doped amplifier, which has an exceptionally high bandwidth in this region, promises an attractive future for optical communication around the 2 μm wavelength. Thus, a viable solution is, therefore, to open a new

optical communication wavelength window around the 2 μm region.

In this article, switches reported are aimed for operation at the 2.1 μm wavelength, using GST embedded within the SOI waveguide. The wavelength selection is inspired by the fact that GST has the lowest attenuation coefficient, $k_{\text{am}} = 0.006$ at the 2.1 μm wavelength around the 2 μm window. The GST material is embedded within the etched Si waveguide to provide a high overlap between the optical mode in the waveguide and the GST.

3.4 Basic Structure of the switch

The thrust of this report is to combine electrical control with optical waveguiding by embedding GST between silicon waveguide strips to provide two useful transmission states for the channel waveguide. Those two self-sustaining states correspond to the amorphous and crystalline phases of GST. The N-type doping for Si beneath the GST proposed here provides adequate electrical conductivity for Joule-heated phase change but does not significantly affect the optical transparency of the waveguide. The SOI approach is selected to make this structure easy to manufacture and compatible with the silicon “CMOS photonics” industry.

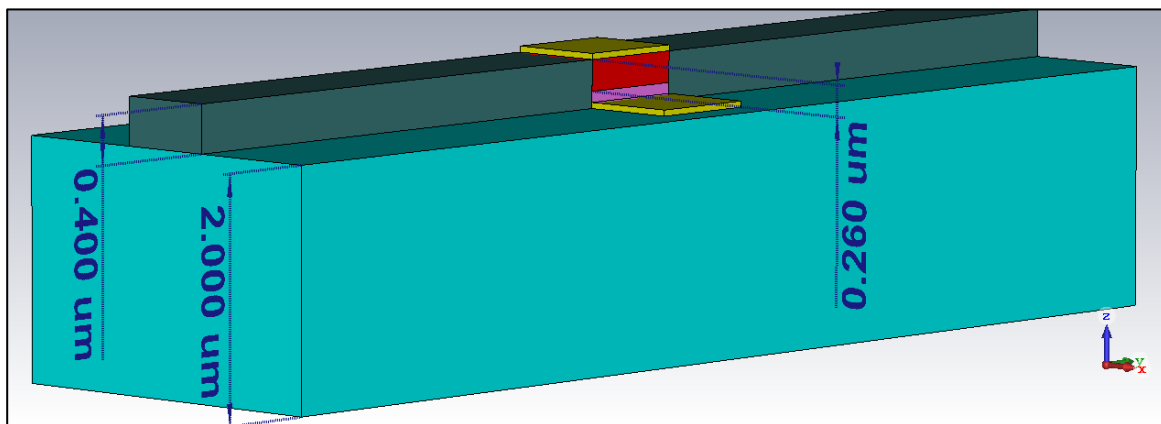


Fig. 1 – Model showing dimensions of the switch

From a theoretical standpoint, the proposed structure in Fig .1 can work properly. In each of its phases, the GST has a complex index of refraction, and the optical waveguide performance is linked to a change in both the real and imaginary indices. We can quantify optical performance using some definitions of indices and absorption. The GST refractive indices are written for the amorphous phase as $n_{am} + ik_{am} = n_1 + ik_1$, and for the crystalline phase, $n_{cr} + ik_{cr} = n_2 + ik_2$, while the effective index for channel waveguide is written as $n_{1e} + ik_{1e}$. Looking at the extinction coefficient k , let us denote α as the absorption loss of the waveguide in decibel per micrometer in each state, $\alpha = 4.34(4\pi k/\lambda)$. The Insertion Loss of an amorphous active waveguide of length L is proportional to the product αL , that is $IL \text{ (dB)} = 4.34(4\pi k_1 L/\lambda)$ and the crystalline-phase loss is $4.34(4\pi k_2 L/\lambda)$. The Extinction Ratio is then $ER \text{ (dB)} = (k_2 - k_1)(4.34)(4\pi L/\lambda)$. Before simulations can be started, it is necessary to have reliable values of the complex indices $n + ik$ of GST. Literature sources of n and k have been studied, and the following values have been used for amorphous-GST & crystalline-GST respectively –

$$\mathbf{a\text{-GST} = 4.05 + i0.006}$$

$$\mathbf{c\text{-GST} = 6.80 + i0.40}$$

Partially crystallized or “fractional” intermediate states are not considered; only the amorphous and the rocksalt-structured crystal phases are examined.

4. Simulations

4.1 Designing the switch

For designing a GST-based photonic switch, we take the approach of embedding GST inside the etched silicon-on-insulator (SOI) waveguide. Such an approach improves the light-matter interaction and provides a high extinction ratio with a small active volume. The phase-change material GST embedded into the Si strip waveguide interacts with the propagating mode depending upon the phase of this material. To incorporate GST, Si is removed partially in the middle of the waveguide & GST is deposited in this space to match the height of the actual waveguide. Hence, the light launched from the input port traverses some length initially in the Si waveguide, interacts with GST, and finally traverses a length again in the Si waveguide before reaching the output port. This approach is taken to minimize the interface scattering and reflection losses between the active waveguide region and the adjacent passive waveguide regions. This approach drastically reduces the insertion loss of the device. Because of the different values of GST extinction coefficient, the optical mode suffers variable attenuation in two phases. In the crystalline phase, because of the large value of the extinction coefficient, GST behaves as a lossy waveguide, and the propagating mode suffers significant attenuation as most of the light is absorbed by it. In contrast, the mode propagates with lesser attenuation in the amorphous phase because of the small value of the extinction coefficient than that of the crystalline phase. Therefore, the amorphous and **crystalline** phases of GST correspond to the ON and **OFF** states of the photonic switch, respectively. Taking into consideration the insertion loss and the extinction ratio, the optimized value of GST dimensions (length, width & height) have been used. For the phase

transition of GST, we consider the voltage triggered phase change for switching applications. Hence, electro-thermal co-

simulations are performed for the optimized structure of the switch.

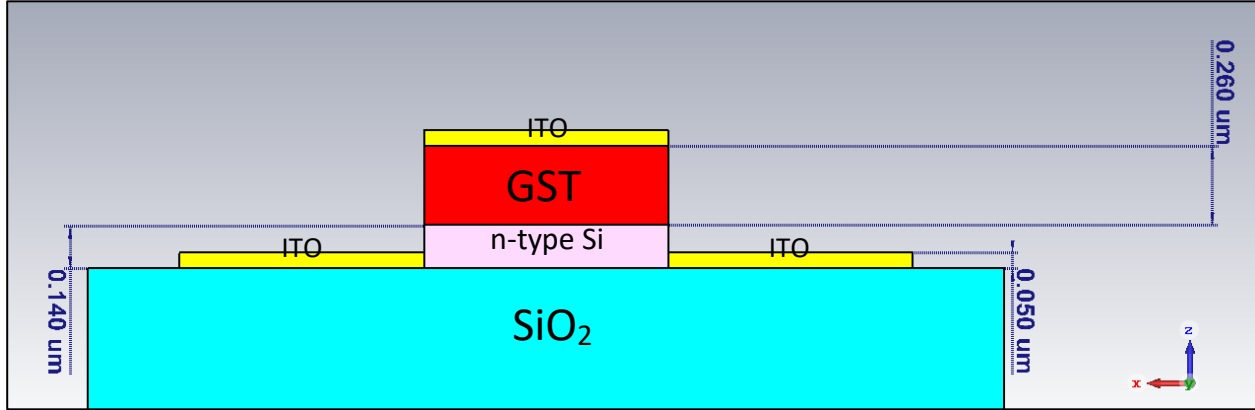


Fig. 2 - Front cross-section view of the proposed 1x1 switch

The switch designed here is a hybrid Si-GST one input-one output (1×1) waveguide switch. The Si strip waveguide is designed in an SOI platform with a **400 nm (height) Si layer** and a 2 μm thick buried silicon dioxide substrate. For single-mode operation at 2.1 μm, the cross-section of the waveguide is **800 nm × 400 nm (width × height)**. The output optical power through the Si waveguide is altered by changing the phase of GST (from a-GST to c-GST and vice-versa) embedded inside the Si waveguide. The optical mode is launched from the left side, and the output is measured from the right side of the waveguide (Fig. 3). Indium-tin-oxide (ITO) electrodes are used to inject voltage pulses into the GST to induce phase transitions. ITO is chosen as the electrode material as its deposition temperature is around 573K (300° C) & this temperature is

CMOS compatible. Therefore ITO is an electrode material that can be introduced into the CMOS process. Each of the **ITO** layers has a height of **50 nm**. To achieve a high extinction ratio and low insertion loss, the GST dimensions were varied. Insertion Loss & Extinction Ratio are defined as -

$$IL = 10\log(T_{ON}), \text{ and } ER = 10\log(T_{ON}/T_{OFF}),$$

where T_{ON} and T_{OFF} denote the transmission for the amorphous and crystalline phases, respectively. H_{GST} and H_{Si} are varied in such a way that the total height of the waveguide remains the same ($H_{GST} + H_{Si} = 400 \text{ nm}$). Based on the best value of ER and IL, the optimal dimensions of GST for the electro-thermal simulations were found to be **920 nm × 800 nm × 260 nm (length x width x height)**.

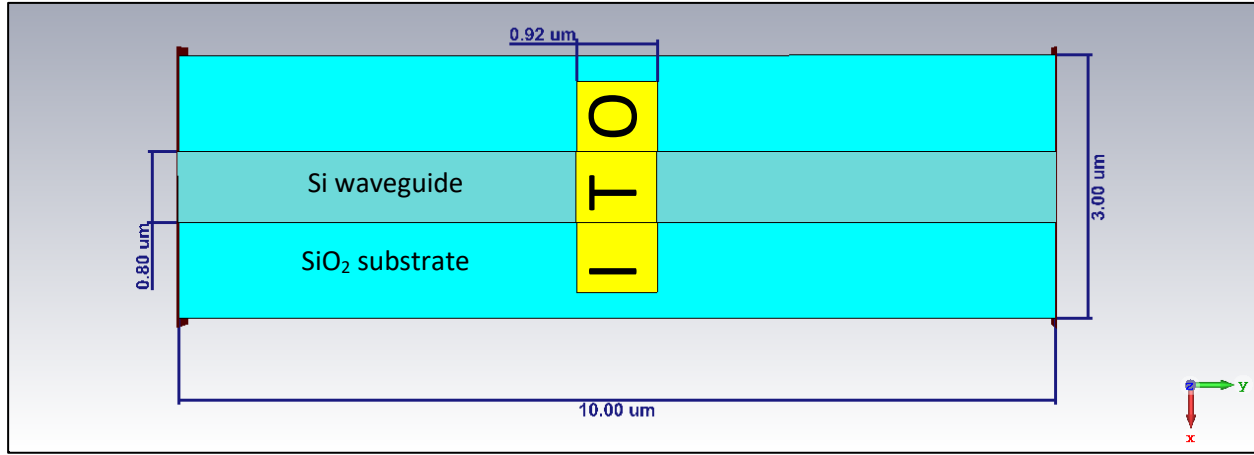


Fig. 3 – Top view of the proposed switch

4.2 Governing Equations

All the simulations are performed using the CST Microwave Studio. The electromagnetic investigations are completed using a three-dimensional finite integration technique with open boundary conditions. The thermal studies are conducted by electro-thermal co-simulations using the M-physics module of this software.

The governing equations for the electrical and thermal simulations are as follows:

$$\nabla \cdot [\sigma(x, y, z, t) \nabla V] = 0 \quad \dots\dots\dots (1)$$

$$\nabla \cdot [k(x, y, z) \nabla T] + Q = C_v \left(\frac{dT}{dt} \right) \quad \dots\dots (2)$$

$$Q = \sigma E^2 \quad \dots\dots\dots (3)$$

In (1), σ and V represent the electrical conductivity of the material and the applied voltage, respectively. In (2), k , T , and C_v denote

the thermal conductivity, temperature, and volumetric heat capacity, respectively.

In (3) Q is the Joule heat generation associated with electric field E .

4.3 Optical Simulations

Optical simulations are performed using the optical module of CST Microwave Studio. This software is based on the finite integration method, and perfectly matched layer boundary conditions are considered for simulations.

The optical properties of the device are modeled, taking the refractive index of SiO_2 , Si, and Air as 1.45, 3.45, and 1, respectively, at 2.1 μm wavelength. For the simulations performed, we define the refractive indices as $n + ik$, where n and k are the real and imaginary refractive indices of the material. For GST, we use the value of refractive index $n_{\text{am}} + k_{\text{am}} = 4.05 + i0.006$ (a-GST) and $n_{\text{cr}} + k_{\text{cr}} = 6.80 + i0.40$ (c-GST) at 2.1 μm wavelength, where n and k denote the real and imaginary parts of the refractive index, respectively. The refractive

index of ITO is set to $1.84 + i0.019$. All the values of the refractive index used are summarized in Table 1 –

MATERIAL	Refractive Index ($n + ik$)	Epsilon (Real)	Epsilon (Imaginary)
a-GST	$4.05 + i(0.006)$	$\epsilon' = 16.402464$	$\epsilon'' = 0.0486$
c-GST	$6.80 + i(0.40)$	$\epsilon' = 46.08$	$\epsilon'' = 5.44$
ITO	$1.84 + i(0.019)$	$\epsilon' = 3.385239$	$\epsilon'' = 0.06992$
SiO ₂	1.45	$\epsilon' = 2.1025$	$\epsilon'' = 0.0$
Si	3.45	$\epsilon' = 11.9025$	$\epsilon'' = 0.0$
n-Si (10^{17})	$3.45 + i(0.0003)$	$\epsilon' = 11.90250009$	$\epsilon'' = 0.00207$

TABLE 1 – Values of refractive indices used for optical simulations

Here epsilon is related to the values of refractive index $n + ik$ as –

$$\epsilon' = n^2 + k^2 \quad \& \quad \epsilon'' = 2nk$$

To demonstrate the working of the designed photonic switch, the absolute value of the electric field profile for the optimized geometry of GST on the partially etched Si waveguide has been plotted in Fig. 4. The refractive index contrast between the amorphous and crystalline phases of GST results in different electric field profiles of the propagating mode and, therefore, different output powers at the end of the waveguide.

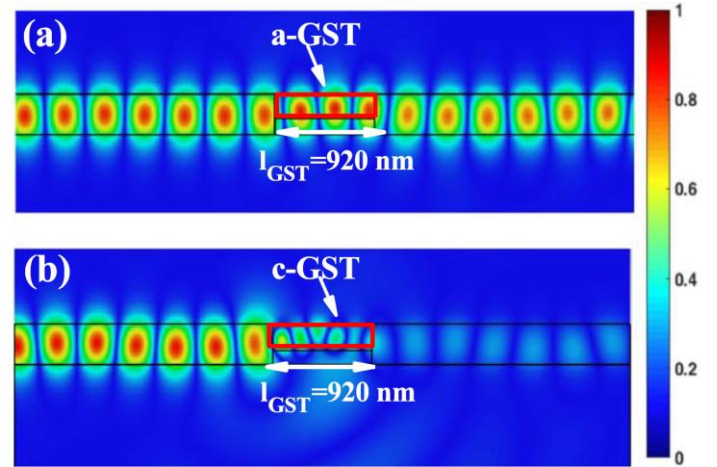


Fig. 4 – Demonstration of the (a) ON state (a-GST) & (b) OFF state (c-GST) [adopted from reference [1]]

In the amorphous phase, Fig. 4 (a), the mode propagates through GST without any appreciable attenuation because of the small value of the extinction coefficient, and thus

with high transmission depicts the ON state of the switch.

In contrast, the crystalline phase, Fig. 4 (b), is highly lossy, and the mode passing through

GST suffers large attenuation due to high absorption. Therefore, with almost negligible transmission, the crystalline phase depicts the OFF state of the switch.

4.4 Thermal Simulations

Thermal simulations are performed using the Multi-Physics module of CST Microwave Studio. Open boundary conditions are used for the simulations. There are two thermal procedures: Crystallization from the amorphous phase to the crystalline phase, and the reverse - Amorphization from the crystalline phase to the amorphous phase.

Relevant parameters (density, specific heat, and thermal conductivity) of SiO₂ and Si are taken from readily available literature. For GST, density, specific heat & thermal conductivity values are taken from the mentioned references. These parameters are listed in Table 2 –

MATERIAL	Density (Kg/m ³)	Specific heat (J/Kg-K)	Thermal conductivity (W/mK)
a-GST	5900	212	0.19
c-GST	6150	212	0.58
ITO	6800	1340	5
SiO ₂	2202	746	1.38
Si	2329	713	140

TABLE 2 – Thermal Properties Used For Electro-Thermal Co-Simulations

For GST, the electrical conductivity depends upon the temperature and its phase. It exhibits a sharp increase in conductivity on transformation from a-GST ($\sigma_{a-GST} = 3 \Omega^{-1} m^{-1}$) to c-GST ($\sigma_{c-GST} = 2770 \Omega^{-1} m^{-1}$).

The phase transition of GST from a-GST to c-GST and vice-versa is induced electrically by heating the GST through Joule heating. For this purpose, we implemented an electro-thermal

co-simulation model using the CST Microwave Studio. In this co-simulation model, the electrical simulations predict the current and voltage distributions in the device, while thermal simulations are used to obtain the temperature distribution in the device. In order to find the voltage and energy required for the complete phase transformation of the GST, the rise in temperature for different sets of applied voltages was examined. The

duration of voltage pulses was then chosen as 100 ns and 10 ns for amorphous to crystalline and crystalline to amorphous phase transformation, respectively. The electrical conductivities used for the materials are summarized in Table 3 –

Material	Electrical conductivity (S/m)
a-GST	3
c-GST	2770
ITO	1.3×10^6
SiO ₂	10^{-15}
Si	3.125×10^{-4}
n-Si (10^{17})	1000

TABLE 3 – Electrical Properties used for Electro-Thermal Co-Simulations

For **crystallization**, the GST layer should be heated **above 413K** (140° C) but **below 819K** (546° C) (the melting point of GST) to induce a complete phase transition. The conductivity of GST in an initial amorphous state is small (0.1 S/m), and without “stimulation,” it is impossible to launch enough current to heat the layer to melt. However, when the electric field component of the applied voltage pulse is higher than a “threshold value,” then the GST becomes conductive with no phase change. To simulate crystallization, we assume that the applied field is above the threshold, that the material starts to crystallize, and that the **a-GST conductivity** has an average value of **3 S/m** during the 100-ns set time. The actual process is far more complicated because the conductivity increases during “set,” and the heating process is accelerated [3]. Using CST

software, we simulated the temperature rise induced by a **5 V set pulse applied for 100 ns** ($E = 50 \text{ V}/\mu\text{m}$). Taking into account the thermal and electrical resistance circuits and the resulting current and electric power dissipation, the modeling yields the spatial distribution and time dependence of temperature rise. Here we expect that the GST temperature rises to 420 K in average during this time interval as required for crystallization.

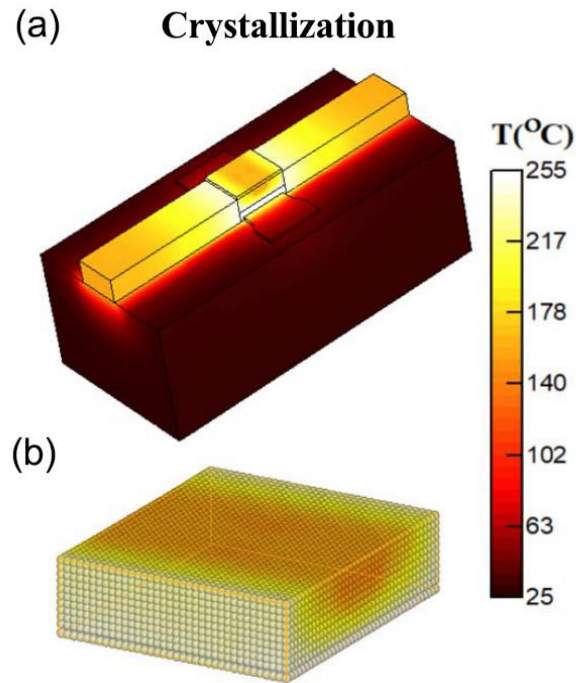


Fig. 5 – (a) Temperature profile of the optimized switch and (b) zoomed view of GST for the process of **crystallization** [adopted from reference [1]]

For **amorphization**, a shorter reset voltage pulse of typically 1 to 10 ns duration is employed, and there the GST layer temperature must be raised above the melting point and then quenched rapidly by the pulse falling to zero in $< 1 \text{ ns}$. In our simulation, a **7.5**

V pulse was applied for **10 ns** to the waveguided **c-GST** having **2770 S/m** conductivity. This produced a very strong temperature rise of ~ 1393 K in the film (above 891 K as desired). In reality, after this “brief” melting, the latent heat of molten GST (1.37×10^5 J/kg) will consume most of the input power, pinning the temperature to the melting point and protecting the contact material. In summary, the electrically switched waveguide is expected to work properly. The voltages that we applied in both simulations might be considered as “relatively high”; however, this is a trade-off we are willing to make to achieve optimum optical performances.

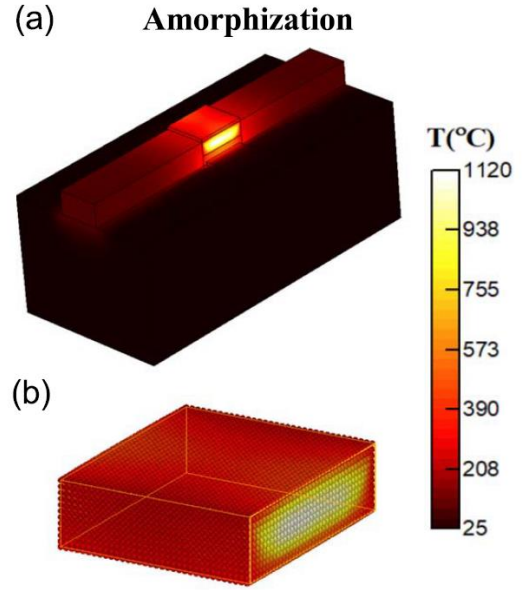


Fig. 6 – (a) Temperature profile of the optimized switch and (b) zoomed view of GST for the process of amorphization [adopted from reference [1]]

5. Observations

The preceding simulations were once again performed for optimized dimensions of GST while varying the dopant concentration in the n-Type doped Silicon. Electrical properties for various levels of doping were easily calculated using $\sigma = q(n\mu_n + p\mu_p)$, where σ is the electrical conductivity, q is the charge on an electron, n is the density of electrons, p is the density of holes, μ_n & μ_p are the respective mobilities. (For more information, refer Solid State Electronic Devices by Ben Streetman).

Optical properties were calculated using the approximations from reference [5].

These are briefly stated here –

$$\Delta\alpha(\lambda) \sim a(\lambda) \Delta N_e^{b(\lambda)}$$

$$-\Delta n(\lambda) = p(\lambda) \Delta N_e^{q(\lambda)}$$

At $\lambda = 2.1 \mu\text{m}$,

$$a = 3.22 \times 10^{-20} \text{ cm}^2; b = 1.149$$

$$p = 1.91 \times 10^{-21} \text{ cm}^3; q = 0.992$$

We use these values to calculate the changes in k and n relative to zero doping case. Hence, we can calculate the new values of ϵ' & ϵ'' as before (using $\epsilon' = n^2 + k^2$ & $\epsilon'' = 2nk$).

MATERIAL	Electrical conductivity (S/m)	Epsilon (Real)	Epsilon (Imaginary)
n-Si (10^{11})	0.002270	$\epsilon' = 11.902499999329$	$\epsilon'' = 1.6170132828020 \times 10^{-11}$
n-Si (10^{12})	0.02269	$\epsilon' = 11.902499993450$	$\epsilon'' = 2.2788429336 \times 10^{-10}$
n-Si (10^{13})	0.2266	$\epsilon' = 11.902499935375$	$\epsilon'' = 3.21154781194 \times 10^{-9}$
n-Si (10^{14})	2.2502	$\epsilon' = 11.902499365551$	$\epsilon'' = 4.52599824867 \times 10^{-8}$
n-Si (10^{15})	21.767	$\epsilon' = 11.902493771320$	$\epsilon'' = 6.37843713459 \times 10^{-7}$
n-Si (10^{16})	188.536	$\epsilon' = 11.902438850137$	$\epsilon'' = 8.98903926626 \times 10^{-6}$
n-Si (10^{17})	1164.8223	$\epsilon' = 11.901899702101$	$\epsilon'' = 1.2502484691 \times 10^{-3}$
n-Si (10^{18})	4547.5216	$\epsilon' = 11.896606987636$	$\epsilon'' = 1.7848710852 \times 10^{-3}$
n-Si (10^{19})	17316.0173	$\epsilon' = 11.844721270839$	$\epsilon'' = 2.50990611695 \times 10^{-2}$
n-Si (10^{20})	102965.4036	$\epsilon' = 11.343856240681$	$\epsilon'' = 3.461187279131 \times 10^{-1}$
n-Si (10^{21})	877192.9824	$\epsilon' = 7.5033128102432$	$\epsilon'' = 3.8263670875755$

TABLE 4 – Electrical & Optical properties of n-Si used for Electro-Thermal Co-Simulations

Many trends were observed in these sets of simulations. The following plots in Fig. 7 to

Fig. 10 summarize the findings of these simulations -

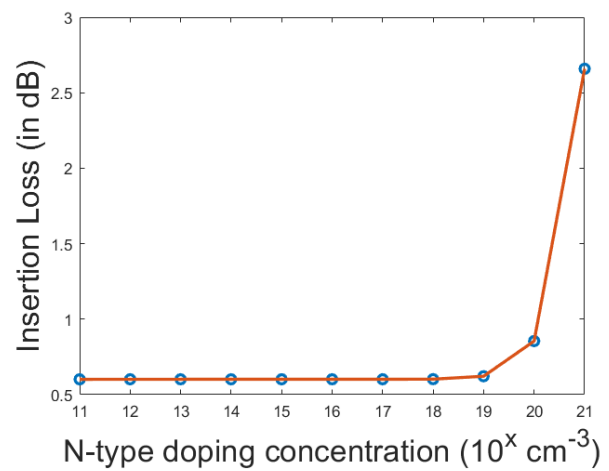
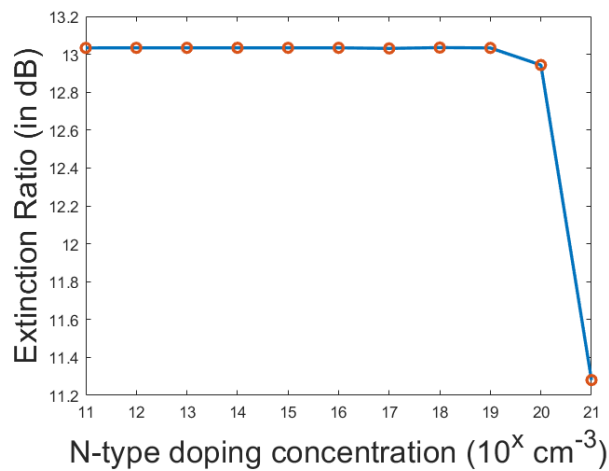


Fig. 7 - Variation of ER & IL for doping concentrations 10^{11} to 10^{21} cm^{-3}

For the 1×1 waveguide switch, Fig. 7 & Fig. 8 show the results for the optimized dimensions of switch design, in which a $0.92 \mu\text{m}$ length GST layer having a cross-section of $800 \times 260 \text{ nm}^2$ (W x H) is embedded in a partially etched Si waveguide. The Si layer beneath the GST is N-type doped with a

varying doping density from 10^{11} cm^{-3} to 10^{21} cm^{-3} to provide adequate electrical conductivity. We can clearly see that **ER & IL remain virtually constant with variation in the concentration of dopant atoms within N-type Silicon**, as expected.

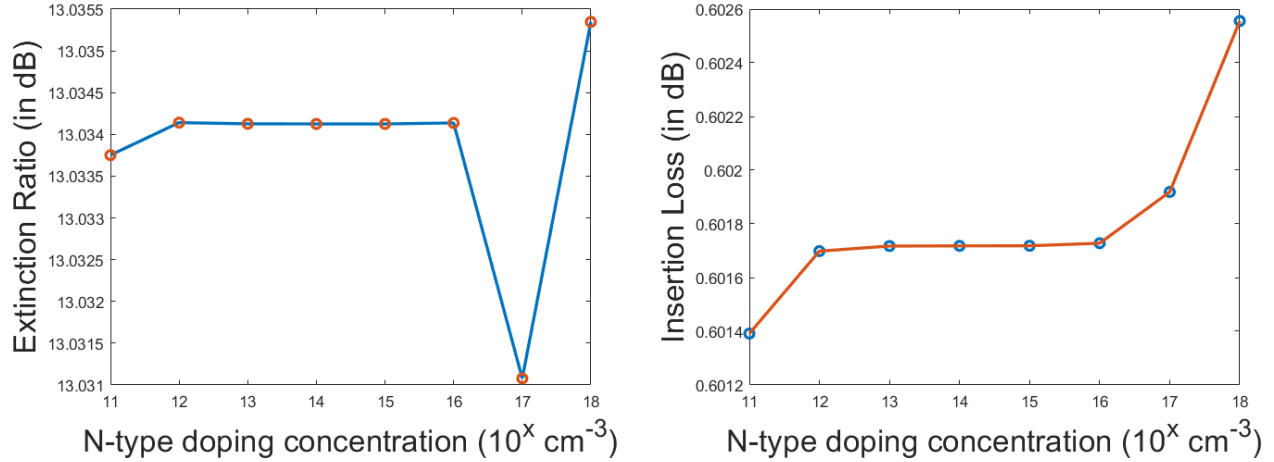


Fig. 8 - Variation of ER & IL for doping concentrations 10^{11} to 10^{18} cm^{-3}

For electro-thermal co-simulations, we plot the maximum temperature attained within the GST embedded in the waveguide. Like before, we use a pulse of 5V for 100 ns for the case of crystallization

and a 7.5 V pulse for 10 ns (falling to zero in $< 1 \text{ ns}$) for amorphization. The **green lines** in Fig. 10 depict the desired range of temperature required for amorphization and crystallization, respectively.

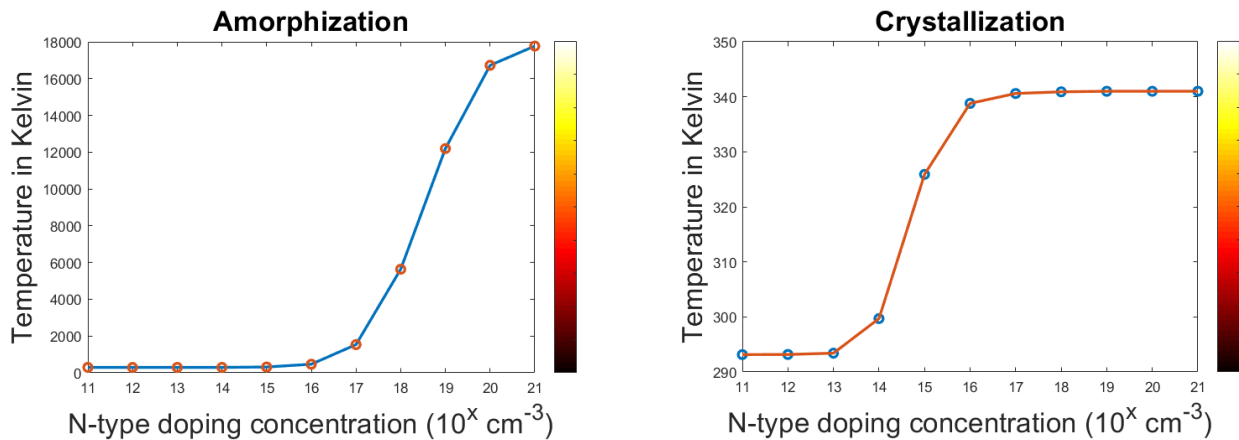


Fig. 9 - Variation of Max Temp. attained for doping concentrations 10^{11} to 10^{21} cm^{-3}

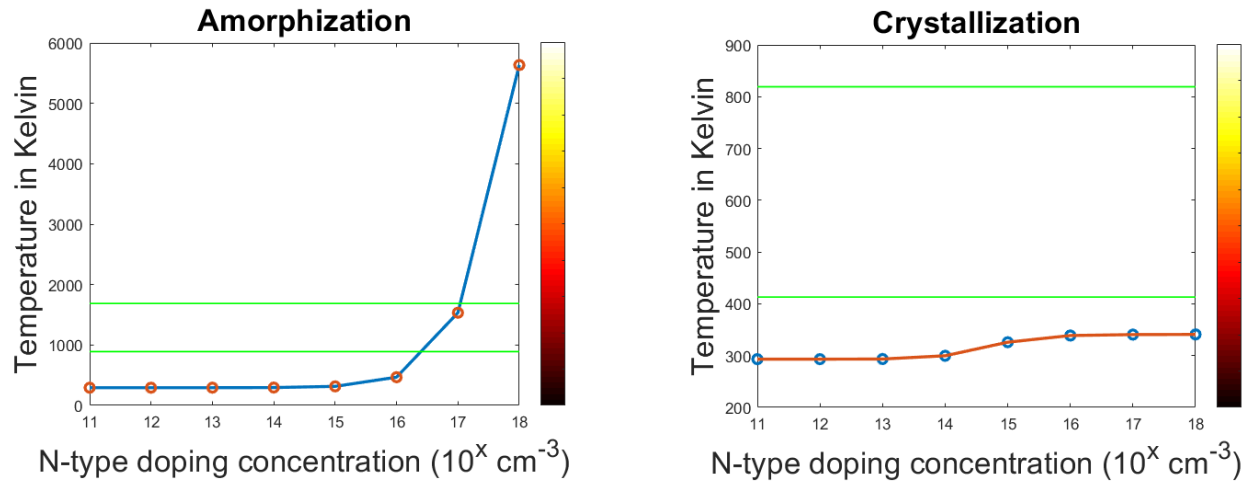


Fig. 10 - Variation of Max Temp. attained for doping concentrations 10^{11} to 10^{18} cm^{-3}

Seeing the results of thermal simulations, we observe that the required temperature is attained in the case of amorphization for the concentration 10^{17} cm^{-3} of dopant atoms. Unfortunately, in the case of crystallization, accurate results, i.e., the required temperature was not attained as reported in reference [1].

Hence, more rigorous simulations need to be performed in this case. However, the trend of temperature variation shows that **the process of amorphization is more sensitive to the concentration of dopant atoms** as compared to the process of crystallization.

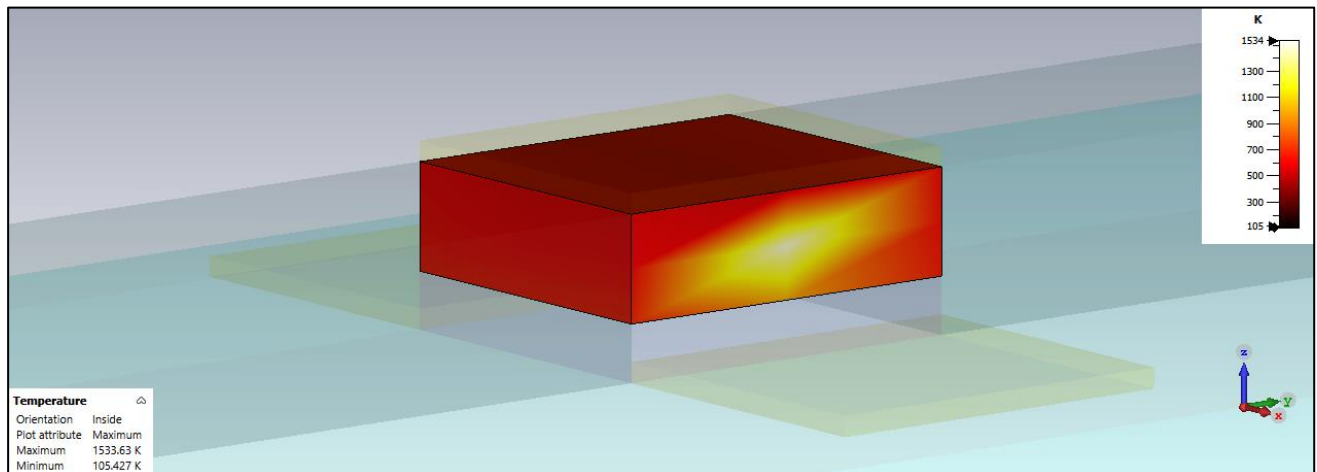


Fig. 11 - Temperature distribution in GST during amorphization for 10^{17} cm^{-3} n-type Si

6. Summary & Conclusion

In conclusion, we have designed and theoretically analyzed an ultra-compact, nonresonant, hybrid Si-GST integrated photonic switch for high static performance in the emerging wavelength window of mid-infrared around 2 μm . The results of this study reveal that variation in doping concentration of n-type Si beneath the GST leads to selective variations in the performance of the photonic switch.

The largest ER value occurs for a very narrow range of GST length. This is due to the combined effect of absorption in GST and reflection and scattering at the interfaces of GST and Si. The maximum ER obtained is **13.0353472 dB** with an **IL of 0.6025558 dB** for 10^{18} dopant atoms per cm^{-3} of n-type Si. These values of ER and IL are obtained for the **GST volume of $920 \text{ nm} \times 800 \text{ nm} \times 260 \text{ nm}$ ($l \times w \times h$)**. Reversible on-off switching is achieved by the electrical actuation of GST through ITO electrodes.

For the process of crystallization, the maximum temperature reached increases with increasing concentration of dopant atoms. For the case of amorphization, the maximum temperature attained is almost constant over the range of doping concentration of n-type Silicon. This effect may be attributed to the fact that GST has an

electrical conductivity of 3 S/m in the amorphous phase & 2770 S/m in the crystalline phase. Moreover, the trends observed in optical & electro-thermal simulation for concentrations above 10^{19} cm^{-3} might be because the approximations for the values of R.I. used are not valid for these concentrations. In addition, we used the same voltage pulse (**5V for 100 ns & 7.5V for 10 ns, in the case of crystallization & amorphization respectively**) for different levels of doping and hence the temperature reaches a very high value for higher concentrations. **We can achieve suitable temperature for higher concentrations using low energy signals.**

Due to fabrication errors, there is a likelihood that the length after fabrication differs from the one that is intended. The proposed switch should be able to maintain the reported ER for a GST length variation of $\pm 40 \text{ nm}$. However, dedicated experimental investigations are needed to verify these results. The work reported here will be useful for the experimental realization of ultra-low footprint, low power-consuming & high on-off ratio PCM-based hybrid integrated EO switches that can be built using GST on a silicon platform for application in integrated circuits for inter-chip and intra-chip photonic applications.



7. References

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